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Flight Program:			
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	When I took over this system, there were two factors related to this system. One was the resolution of the system which was mainly dependent on the signal to noise ratio (SNR) of the detectors and on the power stability of the light source. The other was the stability over time of the measured relative shift which was strongly dependent on the temperature of the photo detectors' circuits. I got two sets of measurements that have good linearity and can show the linear trend line shift because the temperature of one detector has been shifted a little bit. This relation could describe the criterion for temperature controller if we need the accuracy of the original system. Since the mechanical layout and the positions of TEC plate and thermistor have been modified, the system needs to be remeasured in the future. For relative shift data, some data had a bigger difference from the average value. One kind was wrong data. The other was abnormal data. Both kinds of data were irregularly. The waveforms of wrong data were smooth. After I rewrote the relative shift analysis program by LabVIEW to compare with MATLAB source code, I modified the MATLAB code to		

Task Description:

get the correct relative shifts for avoiding the hard job on hardware synchronization. Those waveforms of abnormal data had some asymmetry distribution of noise. To improve the ground problems and EMI problems can reduce this kind of noise. It was necessary to remove the bigger uniform noise before to see the asymmetry noise because the asymmetry distribution of noise was mixed with the bigger uniform noise. In the measurements of eye model, the light is reflected from eye model by Brewster's angle and its intensity is very low compared with a straight alignment system. After increased the gain by changing a resistor and a capacitor on the feedback circuit, the noise was very obvious. The bigger uniform noise was generated by the TEC plate, the thermistor, the current passing through a thermistor from temperature controller, EMI effects and ground problems. After I replaced the temperature controller built by WTC3243 chip with NewPort 325B temperature controller and changed the position of the thermistor with the mechanical layout of the TEC plate, it reduces the amplitude of noise. The EMI effects and ground problems not only generated the uniform noise but also generated the non-uniform noise. After I connected heat sinks and some parts of this system with ground or the optical table and covered the extended cable of photodiode with wire mesh, it reduces uniform noise and non-uniform noise. In the meanwhile, to move any power source or power line as far as possible. In addition to those effects of EMI, the FLUKE Hydra Logger will affect measuring results when the data logger program is running scan mode to download data. It needs other methods to download temperature data from temperature controllers. In the future, I will download temperature data from the USB port of the NewPort temperature controller or from some acquisition board.

The original eye model has the issue for interfering with housing if we need the second spot light to pass into detectors by near the Brewster's angle reflection from the fused silica lens. The view angle for this original eye model is theoretical and the specification of housing didn't concern the angle of marginal ray from light source. I changed the specification of housing under the possibility of mechanical layout modification. Then it can make the second spot light passing into detectors near the Brewster's angle reflection from the fused silica lens without interference.

There are some slow variations or drift issues related to temperature for measurements of the eye model. But it also can take measurement of glucose solution by measuring the de-ionized water as a reference level for the calibration after taking few data of glucose solution. After the calibration of the reference relative shift from de-ionized water, the relation between relative shifts and glucose concentration has a good linearity. Before taking measurements of a human eye, it needs the eye model to verify the stability of this system that is suitable or not for human eye's measurement. In the future, the eye model can verify the alignment method is good or not on human eye's measurement. I will fix the drift issue related to temperature on the next step.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

A non-invasive method of monitoring blood glucose level through the aqueous humor in the eye using the principle of optical polarimetry and Brewster reflection has been explored by Dr. Rafat R. Ansari from NASA Glenn Research Center. This report covers the research and development effort performed to further advance the implementation of the method proposed by Dr. Ansari.
*Measurements on straight-aligned system

For the 10 mm light path of cuvette, the theoretical relative rotation of light polarization for two wavelengths (500nm and 700nm) on 100 mg/dl difference is 3.95 milli-degrees. On this system, if the temperatures on both detectors are stable, the standard deviations from 50 measurements are distributed usually between 0.1 milli-degree and 0.3 milli-degree. The square of the correlation coefficient for relative phase versus glucose concentration measurement for straight-aligned system can better than 0.9965. For 0.1 milli-degree standard deviation on measurement, the concentration accuracy is about 3 mg/dl. According to measurements of straight-aligned system, this system needs the stability of temperature better than 0.001°C if it needs the standard deviation is 0.01 milli-degree on this system. The mechanical layout and the positions of TEC plate and thermistor have been modified. The system needs to be remeasured in the future.

*Software inspection and modification

Some wrong data happened irregularly. The waveforms of such wrong data were smooth. Since the acquisition starter position affects both angles and signs, it needs to synchronize the starter position. But it's very difficult because the gap time for each continuous acquisition is not a constant. And the starter angle is not a constant for continuous acquisition. After I rewrote the relative shift analysis program by LabVIEW to compare with MATLAB source code, I modified the MATLAB code to get the correct relative shift that wrong data disappear after re-calculating the raw data by shifting the initial phase angle to integrate.

*Reduce uniform noise

In the measurements of eye model, the light is reflected from eye model by Brewster's angle and its intensity is very low compared with a straight alignment system. After increasing the gain by changing a resistor and a capacitor on the feedback circuit, the noise was very obvious. In this system, the noise can be separated by two kinds. One is uniform noise. The other one is affected by something changed. The original temperature control was built by WTC3243 chip of Wavelength Electronics, Inc. I found the constant current passing a thermistor near components of the circuit board will generate the noise to this system. To replace the WTC3243 temperature controller with NewPort 325B temperature controller can reduce the noise from the thermistor. The TEC plate attached behind the circuit board also increased the noise because there was a big current through the TEC plate. After modifying mechanical design, the noise was reduced obviously. Because the TEC plate and thermistor affected the measuring signal, it should consider the relation between temperature and relative shift again in the future.

*Reduce non-uniform noise

There were some non-uniform noises inside the system generated by ground problems and EMI (Electromagnetic Interference) problems. There are some other abnormal data happened in this system and it's hard to see the difference on raw data. Usually the abnormal data will be generated by the asymmetry distribution of noise signal. If the

	background noise is very uniform, it doesn't change too much on phase shift calculation. To improve the ground effect and shield the EMI effect can reduce the noise. To move any power source or power line as far as possible is better for this system. *The data logger interference
Task Progress:	The temperature monitor for a circuit board of the detector usually is necessary because there is an issue between relative phase shifts and temperatures of circuit boards. The data logger interference doesn't appear on measurements of straight-aligned system. When the gain was increased for measurements of the eye model, many small effects were amplified obviously. On large gain for this circuit, it shows that the EMI (electromagnetic interface) generated from FLUKE Hydra Logger will affect measuring results when the data logger program is running scan mode to download data. It needs other methods to download temperature data from temperature controllers. In the future, I will download temperature data from the USB port of the NewPort temperature controller or from some acquisition board. *The electromagnetic interference of extended cables for the photodiodes When the human body or power source approached the extended cable for a photodiode, it affected signals of the detector. It also changed the average and standard deviation of relative shifts. For reducing the EMI effect, it needs the shield on extended cables for photodiodes. But when the shielding mesh was added, it would increase the noise of signals. On this system, those unconnected wires of thermistors or TEC plates should be connected with optical table. Then the noise is reduced. *The eye model issue The Brewster's angle from water to fused silica lens is about 47.6°. The angle of incident light from air to contact lens (Etafilcon) forming a Brewster's angle from water to fused silica lens is about 58.3°. The half view angle for original eye model is 60°. This specification is for theoretical condition and can not work on this system correctly because only the first spot has the possibility to pass into detectors close to incident angle 58°. It will interfere with the housing when the focusing light spot is passing through the eye model if we want to use second spot. To interfere with housing will generate a fake spot sometimes. For this tungsten-halogen light source which is connected with one fiber collimator and combined with one lens (EFL=20mm), the marginal ray from focusing lens to minimum spot is about 18°. So, it must modify the view angle of this eye model with half view angle 76.3° at least theoretically. To consider the possibility of mechanical layout modification, the eye model was modified to be half view angle 75°. Then it can make the second spot light passing through RLP by near the Brewster's angle from water to fused silica lens in the eye model. *Measurements on eye model On the measurement of eye model, there is a drift phenomenon for this system. The relative shift of de-ionized water will change gradually. For the long time measurement on de-ionized water, there are two different kinds of this drift issue. One is the global drift. Its trend will increase or decrease gradually affected by optical alignment. The other drift is the local variation on relative shift. Those issues maybe were affected by temperature or others. These issues are more obvious than in straight-aligned system. The global drift issue on straight-aligned system is about below 0.2 milli-degree for relative shift during 1 to 3 hours measurement. But the global drift issue for eye model is about 5 to 8 milli-degrees during 1 to 3 hours measurement. To reduce the drift issues is important in the future. The eye model can verify the stability of this system that is suitable or not for human eye's measurement. In the future, the eye model can verify the alignment method is good or not on human eye's measurement. One this alignment, the light intensity of second spot from the eye model near the Brewster's angle reflection usually is 0.1% of the incident light intensity. For this slow variation or drift issue, it also can take measurement of glucose solution. After taking few data of glucose solution, measure the de-ionized water again. Take the relative shift of de-ionized water as a reference to calibrate the phase shift of glucose solution. The standard deviation of eye model measurement is between 3 and 4 milli-degrees in repeated 50 times for statistical purposes. The square of correlation coefficient can be 0.9524. The higher light intensity of light source we used, the lower standard deviation of measurement we got.
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